On my honor, I have neither given nor received any unauthorized and/or inappropriate assistance for all sessions of this exam. The work done on this exam totally my own. I understand that by the school code violation of these principles will lead to a zero grade and is subject to hash discipline issues"

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1. consider the matrix $A = \begin{bmatrix} 1 & \alpha & -4 & \alpha \\ 3\alpha & 5 & -9 & 1 \\ 2 & \beta & 3 & \alpha^2 \\ 1 & 14 & -6 & 2 \end{bmatrix}$

a. Find the unkown entries α, γ, β, k and m ?
b. Find the eliminated upper triangular matrix U ?
c. Determine remaining elimination matrices.

a. k and m

$k x_2 + x_4 = 0$

$x_2 = x_3$

$-\frac{1}{2} x_3 + x_4 = 0$

$m x_1 + x_3 = 0$

$m_1 x_1 = -x_3$

$x_2 = x_3$

$k = -\frac{1}{2}$

$m = -8$

$-8 x_1 + x_2 = 0$

$x_2 = 8 x_1$

$x_2 = 8 x_1$ and $m x_1 = x_3$

$m = -8$

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3. Question

a. $H^2 = H.H$ and given $H = I - 2uu^T$

H^2 also equals to $(I - 2uu^T)^2 = (I-2uu^T).(I-2uu^T)$

$= I^2 - I\,2uu^T - 2uu^T.I + 4uu^T uu^T = I(I - 2uu^T - 2uu^T) + 4u\|u\|^2 u^T$

$\|u\| = 1$ and $\|u\|^2 = u^T u$

$H^2 = I - 4uu^T + 4uu^T$

$\boxed{H^2 = I}$

b. $H.H^T = (I - 2uu^T)(I - 2uu^T)^T$

$= (I-2uu^T)(I^T - 2u^T (u^T)^T) = (I-2uu^T).(I-2uu^T)$

$= I^2 - 2uu^T - 2uu^T + 4uu^T.uu^T$

$= I - 4uu^T + 4u\|u^2\|u^T$

$= I - 4uu^T + 4uu^T$

$= I - 4(H) + 4(H) \Rightarrow HH^T = I$

c. For A $\boxed{Av = \lambda v}$

$Hu = (I - 2uu^T)u$

$= Iu - 2uu^T u$

$= u - 2u\|u\|^2$

$I.u = 1u$ and $\|u\| = +\sqrt{u^T u}$

so, $\|u\| = 1 = +\sqrt{u^T u}$

$u^T u - \|u\|^2 = 1$

With this way

$Hu = u - 2u\,1 = -u$

$Hu = -u - (-1) = u$

We have $Hu = \lambda u$ and $\lambda = -1$

u is an eigen vector and corresponding eigenvalue u is $\lambda = -1$

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d. Given, $H = (I - 2uu^T)$ and $H^T = I^T - (2u^T(u^T)^T)$

$= I - 2(u^T)^T u^T$

$I = 2uu^T - H$

H symmetric matrix.

A matrix orthogonal diagonalizes respectively

$H = A \Lambda A^T$ H diagonalizable matrix.

$= A \Lambda A^{-1}$

$(A^{-1})^{-1} \Lambda A^{-1}$

$P^{-1} \Lambda R$

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4- $v(0)=10$ and $w(0)=40$

a. 1rd dif equation express the proceat because it should be continious.

$$\frac{d}{dt}\begin{bmatrix} v \\ w \end{bmatrix} = \underbrace{\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}}_{0 \quad 0} \leftarrow \text{system eleveizli}$$

$$|A - I\lambda| \Rightarrow \begin{vmatrix} -1-\lambda & 1 \\ 1 & -1-\lambda \end{vmatrix} \qquad \lambda_1 + \lambda_2 = -2 \qquad \lambda_2 = -2$$

$(-1-\lambda)(-1-\lambda)$ ⇐ equation

$$\lambda_1 = 0 \Rightarrow \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} x_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad \lambda_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \text{ eigen vector}$$

eigen volve

$$\lambda_2 = -2 \Rightarrow \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} x_2 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad x_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \text{ eigen vector.}$$

$$v(t) = s_1 e^{\lambda_1 t} c_1 + s_2 e^{\lambda_2 t} c_2$$

$$v(t) = \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{0t} c_1 + \begin{bmatrix} -1 \\ -1 \end{bmatrix} e^{-2t} c_2$$

$$v(t) = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \cdot e^{0t} \cdot c_1 + \begin{bmatrix} -1 \\ -1 \end{bmatrix} e^{-2t} \cdot c_2$$

A tank = 10L

B tank 40L

$$\begin{bmatrix} 10 \\ 40 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} c_1 + \begin{bmatrix} -1 \\ -1 \end{bmatrix} c_2$$

$$10 = c_1 + c_2$$
$$+\ 40 = c_1 - c_2$$

$$c_1 = 25 \qquad c_2 = -15$$

$$v(t) = \begin{bmatrix} 1 \\ 1 \end{bmatrix} c_1 25 + \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{-2t} (-15)$$

$$v(1) = \begin{bmatrix} 1 \\ 1 \end{bmatrix} 2.5 + \begin{bmatrix} -1 \\ -1 \end{bmatrix} e^{-2} (-15)$$

$25 + (-15)e^{-2}$ ⇒ for A ⇒ %45

$25 + 15e^{-2}$ ⇒ for B ⇒ %55

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$V_1 + 1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} 25 + \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{\phi} (-15)$

$t \to \infty$

1.25 A

1.25 B

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